

INVISIBLE NETWORK

	Capcom:	3, 2, 1, 0.
00 00 03	Glenn:	Roger. The clock is operating. We're underway.
00 00 07	Capcom:	Hear loud and clear.
00 00 08	Glenn:	Roger. We're programming in roll okay . . . Little bumpy along about here.
00 00 16	Capcom:	Roger. Standby for 20 seconds.
00 00 19	Glenn:	Roger.
00 00 20	Capcom:	2, 1, mark.
00 00 23	Glenn:	Roger. Backup clock is started . . . Fuel 102-101, oxygen 78-100, amps 27.
00 00 39	Capcom:	Roger. Loud and clear. Flight path is good, 69.
00 00 43	Glenn:	Roger. Checks okay. Mine was 70 on your mark . . . Have some vibration area coming up here now.
00 00 52	Capcom:	Roger. Reading you loud and clear.
00 00 55	Glenn:	Roger. Coming into high Q a little bit; and a little contrail went by the window or something there.
00 01 00	Capcom:	Roger.

00 01 03	Glenn:	Fuel 102-101, oxygen 78-101, amps 24. Still okay. We're smoothing out some now, getting out of the vibration area.
00 01 16	Capcom:	Roger. You're through max Q. Your flight path is . . .
00 01 19	Glenn:	Roger. Feels good, through max Q and smoothing out real fine . . . Cabin pressure coming down by 7.0; okay; flight very smooth now . . . Sky looking very dark outside . . . Cabin pressure is holding at 6.1 okay.
00 01 46	Capcom:	Roger. Cabin pressure holding at 6.1.
00 01 49	Glenn:	Roger. Have had some oscillations, but they seem to be damping out okay now. Coming up on two minutes, and fuel is 102-101, oxygen 78-102. The g's are building to 6.
00 02 07	Capcom:	Roger. Reading you loud and clear. Flight path looked good. Pitch 25. Standby for . . .
00 02 12	Glenn:	Roger. BECO, back to 1.25 g's. The tower fired; could not see the tower go. I saw the smoke go by the window.
00 02 21	Capcom:	Roger. We confirm staging on TM.
00 02 24	Glenn:	Roger . . . Still have about 1.5 g's. Programming. Over . . . There the tower went right then. Have the tower in sight way out. Could see the tower go. Jettison tower is green.
00 02 48	Capcom:	Roger.
00 02 50	Glenn:	1.5 g's.
00 02 53	Capcom:	Roger, Seven. Still reading you loud and clear. Flight path looks good.
00 02 56	Glenn:	Roger. Auto Retro-Jettison is off; Emergency Retro-Jettison Fuse switch, off; Retro-Jettison Fuse switch, off.
00 03 03	Glenn:	UHF/DF to normal.
00 03 19	Capcom:	Flight path looks good; steering is good.
00 03 22	Glenn:	Roger. Understand everything looks good; g's starting to build again a little bit.
00 03 30	Capcom:	Roger.
00 03 32	Capcom:	Friendship Seven. Bermuda has you.
00 03 34	Glenn:	Roger. Bermuda standby . . . This is Friendship Seven. Fuel 103-101, oxygen 78-100. All voltages above 25, amps 26.
00 03 48	Capcom:	Roger. Still reading you loud and clear. Flight path is very good. Pitch -3.
00 03 53	Glenn:	Roger . . . My pitch checks a -7 on your -3.
00 04 00	Capcom:	Roger, Seven.

00 04 08	Glenn:	Friendship Seven. Fuel 103-101, oxygen 78-100, amps 25, cabin pressure holding at 5.8.
00 04 20	Capcom:	Roger. Reading you loud and clear. Seven, Cape is Go; we're standing by for you.
00 04 25	Glenn:	Roger. Cape is Go and I am Go. Capsule is in good shape. Fuel 103-102, oxygen 78-100, cabin pressure holding steady at 5.8, amps is 26. All systems are Go.
00 04 44	Capcom:	Roger. 20 seconds to SECO.
00 04 47	Glenn:	Roger . . . Indicating 6 g's . . . Say again.
00 04 53	Capcom:	Still looks good.
00 04 54	Glenn:	Roger . . . SECO, posigrades fired okay.
00 05 10	Capcom:	Roger, stand . . .
00 05 12	Glenn:	Roger. Zero-g and I feel fine. Capsule is turning around . . . Oh, that view is tremendous!
00 05 21	Capcom:	Roger. Turnaround has started.
00 05 23	Glenn:	Roger. The capsule is turning around and I can see the booster during turnaround just a couple of hundred yards behind me. It was beautiful.
00 05 30	Capcom:	Roger, Seven. You have a go, at least 7 orbits.
00 05 35	Glenn:	Roger. Understand Go for at least 7 orbits . . . This is Friendship Seven. Can see clear back; a big cloud pattern way back across towards the Cape. Beautiful sight.
00 05 54	Capcom:	Roger, still reading you loud and clear. Next transmission, Bermuda.
00 05 58	Glenn:	Roger. Understand next transmission, Bermuda . . . [1]

Bermuda, Canary Island, Kano, Zanzibar. The list goes on: Muchea, Woomera, Canton Island, Guaymas. Reading like a who's who from National Geographic, these obscure spots from around the world became astronaut John H. Glenn, Jr.'s only links back to Earth on that historic day in 1962 as he circled the globe at 27,350 kilometers per hour (17,000 miles per hour) inside *Friendship 7.* Completing an orbit every 90-minutes, ground stations that the National Aeronautics and Space Administration (NASA) had constructed at these and other places around the world anxiously tracked the progress of Glenn in his tiny, Mercury capsule. Not unlike runners in a relay, as *Friendship 7* passed from station to station, each Capsule Communicator—known as Capcom—handed him off to a waiting colleague down the line.

Even though the momentous flight of 20 February 1962 lasted only 4 hours and 55 minutes, the United States had prepared for it since before NASA even opened its doors as America's space agency some three and a half years earlier. By the time Glenn became the first American in orbit, NASA

had in fact already established no less than 30 ground stations on five continents and several islands, along with ships in the Atlantic, Indian, and Pacific oceans. At the time, about half of these stations were devoted to tracking science and application satellites while the other half played a unique role in the then-burgeoning arena of human spaceflight, an arena which seven years later, would culminate with mankind's first steps on the Moon.

Connected by over two million circuit miles of land and ocean-floor cables, America's spaceflight network spanned just about every corner of the globe, from desolate volcanic atolls like Ascension Island to metropolitan, capital cities like Madrid and Canberra.[2] In the United States, stations could be found spanning the cold north of Alaska to the lush hillsides of North Carolina. Like the deep-sea diver who relies on his harness and cable to communicate with and find his way back to the mother ship, NASA's spaceflight tracking and data network was—and is—that electronic link to the satellites, spacecraft, and astronauts in space.

More than that, it is also an electronic link to the past.

Through the ages, communications have been the key to discovery. Whether by foot messengers or highspeed electronics, communications have been essential to the exploration of new frontiers. Take the 1960s, for example, where radio communication with probes preceded man to the Moon. Regardless of how sophisticated it may be, no spacecraft is of any value unless it can be tracked accurately to determine where it is and how it is performing. Only in doing so can the data it is collecting—whether pictures of celestial objects or television broadcast signals—be transmitted, received and used on the ground. This data, reduced into useful information by computers and electronics on the ground, enable the user here on Earth to analyze data from space. In the case of human spaceflight, the stakes are much higher. A failed communications link potentially compromises not just the success of a mission but also puts the lives of astronauts at greater risk.

These electronic downlinks—called telemetry—carry everything from astronauts' pulse-rate to so-called "housekeeping" data, which give an indication of the health and status of the spacecraft. Conversely, uplinks transmit commands from mission controllers, scientists and engineers up to the spacecraft. These radio frequency or RF links bridge the expanse of space, tying the spacecraft and Earth to each other. Without tracking, telemetry, command, and control, satellites would merely be inanimate objects in space and astronauts would be beyond the reach of the thousands who support their mission back on the ground.[3]

William C. Schneider, NASA's former Associate Administrator for Tracking and Data Systems from 1978 to 1980, commented on the vital role that space communications played, for example, during tense moments on Apollo. "Very few people really understand and appreciate the importance of

reliable tracking and communications to all of NASA's programs, particularly when it comes to manned missions. Sitting in Mission Control in Houston during a lunar flight, there was a communications blackout when the spacecraft was behind the Moon. This also happened to be a period when the spacecraft has to fire its engines to start its descent to the lunar surface. Not having communications during this critical phase made us all feel very nervous and anxious, and the relief, when the capsule was able to confirm that all was well, was almost indescribable."[4]

In the early days of space exploration, prominent engineers, scientists, and astronauts were certainly at the forefront of public attention and acclaim. These professionals in a way became the new Magellans and the new Columbuses, the new breed of modern-day explorers and map makers. Instead of crossing oceans at the mercy of the wind, astronauts now traveled the vast expanse of space in craft designed by engineers. Whether a mission was the flight of a communications satellite or the exciting journey of an astronaut, *all* were supported by a unique team of men and women on virtually every continent who operated an intricate system of ground stations, computer facilities, and communication centers. These, tied together, made up NASA's Spaceflight Tracking and Data Network (STDN).

John T. Mengel, one of the "founding fathers" of America's spaceflight tracking networks, once commented on this behind-the-scenes criticality:

> True, tracking and data acquisition were support functions, but this support, going back to the early days of the space effort, was critical to success in the competition-charged arena of the sixties and seventies, when the ability of the United States to succeed in the field of space travel and space research was being questioned. We proved it could be done.[5]

To operate a network, teamwork and cooperation were essential. This was true not just for those working at a station on some remote island but also on the much larger, agency-to-agency level. For instance, over the years, cooperation among the triad of NASA, the Department of Defense (DOD) and the Department of State, has been crucial. To NASA, the Air Force, with its launch facilities and worldwide network of radar installations, was a huge asset. Help from the Navy was important too, providing tracking and recovery ships in all three major oceans. However, like most big projects where different organizations have to depend on each other to get the job done, the NASA/DOD cooperative was not without its share of problems. One example was that the space agency's desire to work with the Air Force was tempered sometimes by the fear that the DOD might try to "elbow in" on the fledgling



Edmond C. Buckley was NASA's first Associate Administrator for Tracking and Data Acquisition, leading the new Office of Tracking and Data Acquisition (OTDA) at Headquarters from 1961 to 1968. In this role, Buckley dealt directly with Congress, the DOD and the State Department to oversee the space agency's tracking needs. Prior to becoming the Associate Administrator, Buckley was NASA's Assistant Director for Space Flight Operations. (Rensselaer Polytechnic Institute photograph, copy in Folder 10/1/1 NASA Australian Operations Office, Yarralumla, ACT)

Agency's new, and in many ways, more glamorous programs. Congress, too, was sensitive to the NASA/DOD relationship. Questions about duplication of facilities, for instance, often came up during budget discussions. Influential members on the Hill played the role of watchdogs to make sure that NASA remained true to its civilian charter. Despite constant budget and political battles, Congress, nevertheless, generally understood that NASA and other government offices simply had to work closely together in order for America to succeed in space.

To maintain communications with an orbiting spacecraft, a tracking network has to be global in nature. Therefore, international cooperation is not only important but absolutely necessary in getting stations established at optimal locations around the world. Whether dealing with strong allies like Australia and Great Britain or venturing deep into Africa for the first time, the role of the State Department was indispensable as a facilitator to help the space agency intermediate discussions at the highest levels of government. With this liaison, officials could more effectively manage NASA operations in the face of geopolitical unrest around the world. For example, during the pioneering flights of Project Mercury, the Guaymas tracking station in Mexico often had to be surrounded by troops to protect it against unruly mobs espousing anti-U.S. sentiment.[6]

The different persuasions of different governments sometimes require delicate diplomacy on the part of the United States. In particular, the ability to cater to the sensitivities of diverse cultures is important. NASA has found that the single, best way to accomplish this is to invite foreign nationals to join in the operation of the tracking network, providing them with a sense



The 85-foot tracking antenna which received signals of mankind's first steps on the Moon on 20 July 1969 still operates today at the Canberra Deep Space Communication Complex in Tidbinbilla, Australia. The antenna was moved from its original Honeysuckle Creek site some 30 kilometers (20 miles) away in 1985. Clearly visible are the two large yokes which allow the antenna to be moved in two "X-Y" axes as it tracks a spacecraft. (Photograph by the author)

of ownership to become partners in the space effort. This "nationalizing of the stations" from the early days of the network would leave an indelible legacy around the world that continues even today.

Such international cooperation could in fact make or break a station. Putting a site in Havana, Cuba in the late 1950s presented a volatile situation. Conversely, other places such as Australia—where as many as 10 sites were active during the 1960s—enthusiastically embraced the opportunity to participate in this new frontier, adopting the American space program as their own. Australia's Parkes Observatory was selected by NASA (with great national fanfare), to help receive the Apollo 11 moonwalk telemetry. Indeed, the Canberra Deep Space Communication Complex (CDSCC) on the outskirts of the Australian capital remains active to this day as part of NASA's DSN. In fact, the 85-foot antenna at nearby Honeysuckle Creek (HSK) that actually received video of Neil Armstrong and Buzz Aldrin's first steps on the Moon is still used at the CDSCC in Tidbinbilla, having been moved from

HSK when it closed in 1984. The years have not changed this. International cooperation is just as vital today as it was then, as the space agency routinely works with the Europeans, Russians and Japanese on a diverse range of programs including the International Space Station, Earth science research, and planning of future space communication needs.

As much as the story of NASA's spaceflight tracking networks is about the stations and the technologies, it ultimately boils down to the individuals, the workers in the trenches who made it happen. From Administrators, to Station Directors, to the teams of contractors, all did their jobs because they believed in it and put their hearts into it. Even though space is often associated with "high-tech," it is still the people involved who were movers of the program, who ran the day-to-day operations of the facilities, and who left behind their legacy. This emphasis on people especially characterized the early years of the Agency's tracking networks. There were no notebook computers in 1957 when the first satellites were launched. In fact, there were no digital computers of any kind. Instead of clicking on a "mouse" to retrieve data, technicians would interrogate lines drawn on graph paper from mechanical strip chart recorders. Whether one was the head of a station or just feeding teletype printouts to the engineers, everyone had a job to do.

While those working with the communication networks of today certainly get their share of excitement, those who were around for the early human flights and satellite launches fondly remember the "glory days." As one former official at the Goddard Space Flight Center (GSFC) recalled:

> I didn't geographically know the world then in the detail I do today. Not only was it educational and exciting just to learn the names of all the remote tracking stations, but there was also the excitement of talking to the people at all the different tracking stations and ships.[7]

Former Flight Director Christopher C. Kraft—who wrote the Foreword to Hamish Lindsay's book *Tracking Apollo to the Moon* on the Australians' perspective of working on the Apollo Network—recalled fondly:

> I have a very soft spot in my heart for the network and the people who operated it. These people were as much a part of the success of our efforts as were the flight controllers and the other people at Houston and the Cape. Whatever we did was in large measure dependent on the reliability of the worldwide tracking and communications network. . . . This is truly one of the unsung accomplishments of the space program.[8]

As the network expanded in the 1960s and early 1970s, so did the team of people who ran it. It was during this time that contractors from the aerospace industry, like Bendix Field Engineering Corporation (BFEC), made an indelible mark on the history of the spaceflight network. So close was the relationship that station workers were often called the "badgeless controller" because it didn't matter if they were government civil servants or contractor employees. The two could work side by side and one would not know which was which unless you looked at his badge.[9] For many, working the networks became a way of life. For some, entire careers were spent working in remote places around the world.

An examination of the history of America's spaceflight tracking network is not unlike driving down a long and winding road—with many turns and twists. Much of its early fate was tied to the Cold War and the Space Race borne from it. In the mid-1950s, even before NASA was formally established, the United States began building a network to track what it hoped was to be the world's first artificial satellite. However, that network—called Minitrack—ended up as its first test tracking Sputnik 1, which the former Soviet Union shocked the world with on 4 October 1957.

Minitrack next set the stage for a greatly expanded network called the Satellite Tracking and Data Acquisition Network, or STADAN. Led by the new Goddard Space Flight Center (GSFC) in Greenbelt, Maryland, this much more expansive network soon became the centerpiece of all near-Earth tracking and space communications activity for the United States. The STADAN supported not only NASA's own science and application satellites but also the world's first commercial satellites which were just then being launched, including the COMSAT Early Bird, the world's first electronically active, commercial, communications satellite.[10]

However, even as the sophistication of satellites was rapidly evolving and launches were taking place on a more regular basis, something even more dramatic was about to change the very fabric of the Goddard network. This something was the world of *manned* or *human* spaceflight.

To support this new national priority, engineers at Goddard, along with the Manned Spacecraft Center in Houston, Texas, developed a network that ultimately tracked astronauts to the Moon and back. Initially called the Mercury Space Flight Network, or MSFN (the word "Mercury" would later be replaced with the word "Manned"), it quickly became just as expansive as the STADAN, even surpassing the latter in terms of capability, locations and new technology. Much of its early requirements were based on experience gained from aircraft and missile testing and from the tracking of lunar and planetary probes such as the unmanned Surveyor spacecraft which preceded man to the Moon. From this, cadres of network planners worked out solutions to the high data rate requirements needed to send humans into space.

As both the STADAN and the MSFN matured, the ever increasing demand for data in the right amount (bandwidth), at sufficient speed (data rate) and at the right time (timing accuracy) drove their evolution. Real time operations at ever-increasing data rate became the prime consideration. "To know accurately where the spacecraft was at any given moment was extremely important to us," said Kraft. "A review of the early history, particularly of the Mercury flights, reveals that we often had to make some very delicate decisions whether to continue some of these flights or not. Determining the orbital trajectory and providing tracking and communications support to meet these parameters from launch to reentry was certainly no easy job for the Goddard team."[11]

To this end, Kraft recalled a specific problem that the network had to deal with involving the critical "Go/No Go" decision as whether to continue a flight or initiate an abort. During the launch of an Atlas rocket—an Air Force Intercontinental Ballistic Missile which NASA used to launch the Mercury astronauts and the Agency's early, large, satellites—there was only a 30- to 120-second window after its main engine had cutoff to decide whether to continue or to abort. This was all the time controllers (or the astronaut) had to turn the spacecraft around and fire its retrorockets so that its trajectory would allow the spacecraft to reenter in the Atlantic recovery area before its impact point reached the African coast. Kraft recalled:

> Initially, there were very few people who believed that this would be possible. . . . We called it the 'short arc' solution. People like John Mayer, Bill Tindall, Lynwood Dunseith and Goddard's Jim Donegan deserve credit for giving us the know-how to solve this critical problem, a problem which many mathematical and trajectory experts believed could not be solved. By analyzing the data generated by the Cape, Goddard and the Bermuda tracking station, we were able to get the answers, even within the very limited timeframe available.[12]

During NASA's first man-in-space program Project Mercury, the network was not well centralized and relied on the not always reliable teletype communications. For mission assurance, flight controllers were dispatched to most of the primary tracking stations so that they could maintain immediate contact with the spacecraft from the ground on a given pass. Astronauts too were sent as capsule communicators or Capcoms to the various sites. Here, something interesting happened within the NASA culture. For unmanned science and application satellites, there was no debate as to who was in charge. Goddard was clearly the lead and the STADAN was centralized at the Network Operations Control Center (NOCC) in Greenbelt. However, things were not

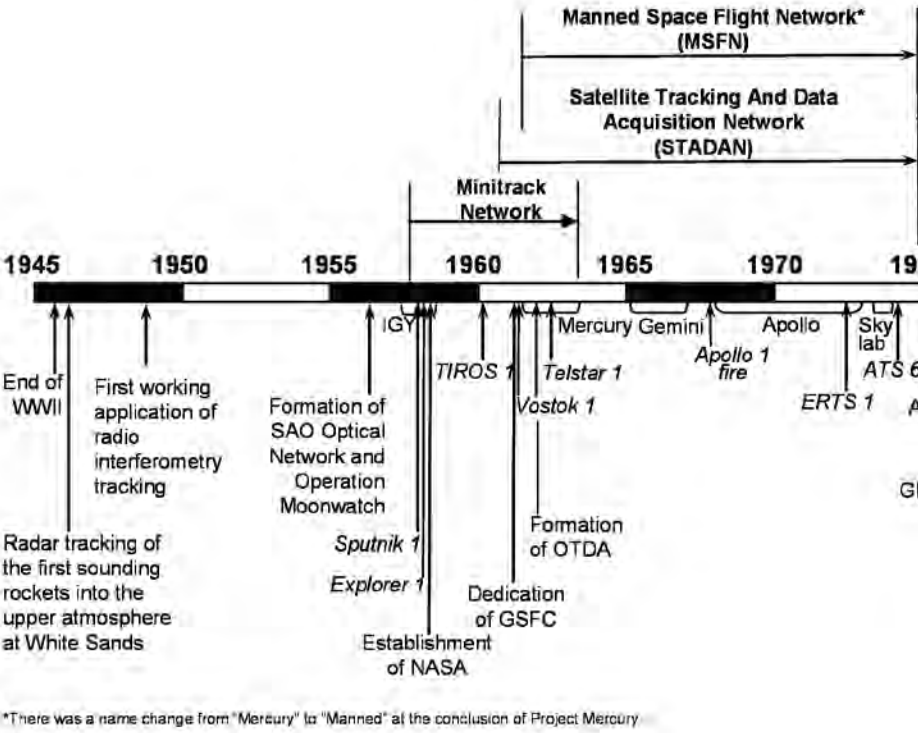
so clear-cut when it came to the "manned" side of the house. Here, both Goddard and Houston's Manned Spacecraft Center (MSC) had a stake.

Despite the fact that there was not always full agreement even within Goddard's own top management (who thought the center should be devoted to scientific exploration of space and questioned whether it should be in the business of tracking human missions), Headquarters in due time delegated Goddard with the full responsibility for running the MSFN. This decision by Headquarters was fairly logical as MSC (renamed the Lyndon B. Johnson Space Center, or JSC, in 1973) certainly had enough on its plate as NASA's lead center for human spaceflight to be worried about also having to run a worldwide tracking network. Nevertheless, the decision did not eliminate Houston's concerns about the network; the "Not Invented Here" mindset was not something that easily went away.

As history has shown, JSC did not have to worry about the Goddard-run network, as not a single mission—manned or unmanned—has ever been compromised due to network failure. This "invisible network" as



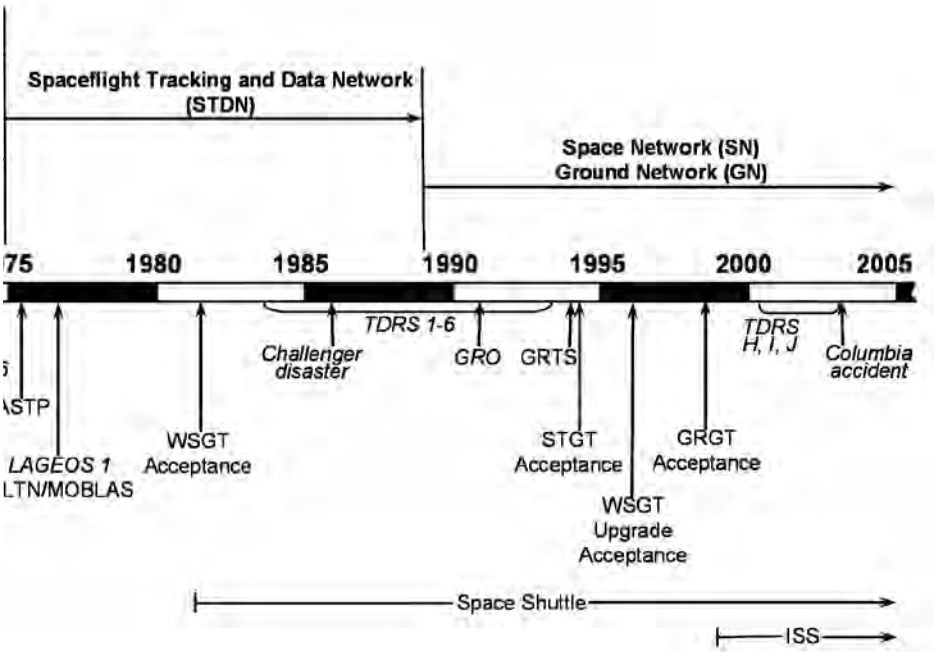
Artist rendition of a Tracking and Data Relay Satellite providing communications support from Earth orbit. The long and winding road has taken NASA's Spaceflight Tracking And Data Network from a handful of ground stations in the late 1950s to the Space Network of the twenty-first century. (NASA Lithograph LG-2001-8-033-GSFC)



A chronology of NASA's near-Earth Spaceflight Tracking and Data Network. (Figure by the author)

it has been called, operating behind the scenes, has always been there. Like a light switch, it's there when needed. In the final analysis, much of the success on what could have been a very divisive issue was due in no small part to some very effective personal relationships between people who understood what was needed to get the job done at both centers, people like Ozzie Covington, Bill Wood, Tecwyn Roberts, and Chris Kraft.

Following the incredible journeys of Apollo to the Moon and America's first space station Skylab, as well as the world's first international spaceflight in the Apollo-Soyuz Test Project, human spaceflight became suddenly very quiet in the late 1970s. Although the GSFC was busy supporting Earth science and application satellites, in the cost-conscious days following Apollo, the wisdom for having a separate network specifically for human spaceflight was questioned. It was in this era of change that Goddard merged its two networks (the STADAN and the MSFN) in 1975 into a single, leaner, and ostensibly more cost-effective network called the Spaceflight Tracking and Data Network, or STDN.



But no matter how large or small a ground network is, due to "line-of-sight" limitations imposed by the curvature of Earth, it is still located on the surface of Earth and can thus only provide coverage for some 15 percent of an Earth-orbiting spacecraft's ground track—limited to brief periods when it is within the line-of-sight over a given tracking station. To overcome this, NASA's network for tracking and communicating with near-Earth spacecraft (that is, the STDN but not the DSN) had to move into space. Thus, the Tracking and Data Relay Satellite System, or TDRSS, was born.

With the advent of TDRSS in the 1980s, America's spaceflight tracking and communications network evolved into a full-fledged Space Network (SN) constellation of relay satellites and ground terminals along with a greatly reduced Ground Network (GN). Today, many Earth-orbiting spacecraft rely primarily on the SN for their tracking, telemetry and control needs with many missions that do not require continuous contact with the ground opting, mainly for cost reasons, to transmit their spacecraft science data via the GN. Digital and telecommunications technology have also evolved significantly in recent

years to impact space communications. The ever increasing demand for higher bandwidths (traffic capability) and lower "bit-error rates" (higher accuracy) in the modern information age explosion have led to the complete transformation of how space communications are done. Such is the trend in spaceflight communications that started in the 1990s in which NASA historically set the precedence but is now heavily influenced by the commercial sector. In this new age of information technology and international teamwork, the Agency's role is no longer the same as it was when the Space Shuttle first flew into space in 1981 or even when the International Space Station was first being built in the 1990s.

Aside from geopolitics, the root cause driving such change can almost always be traced to two factors: technology and cost reduction. Few would be surprised that the demand for better technology is always a driver. But in addition to better technology, as space moves from the realm of government sponsorship to a commercial commodity, cost reduction in today's world of real-time global communications and data demands is perhaps more important than ever before.

In the 1990's, "Faster, Better, Cheaper" also drove much of the way business was conducted in the space industry, both commercially and at NASA. This approach impacted the space program in ways ranging from economics to performance and, some would say, safety. New ground and space communication networks such as Universal Space Network (USN) entered the playing field. This network provided multi-mission ground terminals which offered users the advantage of low cost services: pay only for what you use. They targeted not only commercial users of satellite services but also government users, including NASA. As we broach the twenty-first century, satellite users now routinely rely on the internet for data access and file transfers. This access extends to such remote locations as the South Pole, where NASA was instrumental in providing internet access to researchers.

When one speaks of space exploration, "high technology" usually comes to mind. To this end, the aphorism "If NASA can put a man on the Moon . . . " has become a part of the English vernacular, perhaps even to the point of being somewhat trite. Satellite and spacecraft—some human, some robotic—sent on their missions into space, are there to make the unknown known. But whatever data that astronauts and instruments record, the information is of no value if it cannot be returned to Earth. At its height, over 6,000 men and women—from network engineers to field technicians to NASA managers—were committed to this vast undertaking. As former Station Director and later Associate Administrator for Space Communications (1989 to 1996) Charles T. Force says of this invisible network, "The nation was well served."[13]

This is the story of that invisible network.